

DLX Microprocessor: FPGA to Physical Design

Our team will implement the physical layout and design of a 5-stage MIPS-style microprocessor, previously designed by the team for Dr. Phillip's Advanced Reconfigurable Computing class (ECE 6740). The design was done in VHDL and targeted for an Altera FPGA. For this project we will use Cadence Innovus for the physical implementation flow. This project involves converting the existing VHDL design into a format compatible with Innovus and generating a final physical layout design with proper placement and routing.

Key project aspects include:

Translation of VHDL design to Innovus-compatible format

Handling of multiple clock domains

Adaptation of Intel IP blocks (PLL, FIFO, RAM, ROM)

Complete physical design implementation using Cadence Innovus

Team Members:

Nick Waddoups

Jared Tobler

Project Tasks and Timeline:*Week 1 - Translation into Innovus-compatible Format*

Research methods for compiling code from Intel's Quartus Prime software into a format that we can use in Cadence Innovus.

Refresh ourselves on the VHDL design and prepare documentation for our report and presentation.

Draft the midterm progress report.

Week 2 - Continued Project Translation into Innovus-compatible Format

Either compile Intel IP blocks into code that Innovus can use, or recreate the IP blocks to get the functionality we need.

Determine clock, I/O, and RAM setup for our final design. (Do we want to use on-chip RAM? What do we want for our I/O pins? How are we handling the multiple clock domains?)

Submit midterm progress report.

Begin drafting project presentation slides and final project report.

Week 3 - Initial Setup and Runs in Innovus

Setup basic placement and routing runs in Innovus for our design. Put together build scripts.

Determine where our issues will lay, and prepare to adjust design to resolve issues.

Continue working on project presentation slides and final report.

Week 4 - Presentation and Final Runs in Innovus

Fix issues discovered in Innovus last week.

Finalize and submit project presentation slides and final report.

Project Deliverables:

Converted design files compatible with Cadence Innovus

Custom implementations or equivalents for Intel IP blocks

Complete set of design constraints

Final physical layout with placement and routing results

Timing, power, and area analysis reports

Project documentation and presentation materials

This implementation project will provide valuable hands-on experience with advanced physical design tools and methodologies while demonstrating our understanding of processor architecture and VLSI design principles.